Signal-to-Shock

Financial event intelligence and portfolio stress testing

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The analyst’s decision

Which positions should this headline affect?

A bank default can affect its equity, loan and swap. Applying the same issuer shock to an unrelated technology company distorts the scenario.

Our approach

Show the matched evidence.
Let the analyst choose scope.
Explain every position’s P&L.

Designed for event triage and reproducible scenario discussion.
The shocks are illustrative assumptions.

System design and data flow

INPUT

Synthetic text
Portfolio CSV

RULE ENGINE

Sentiment counts
Event evidence

ANALYST REVIEW

Scenario + issuer
Intensity 1–10

STRESS ENGINE

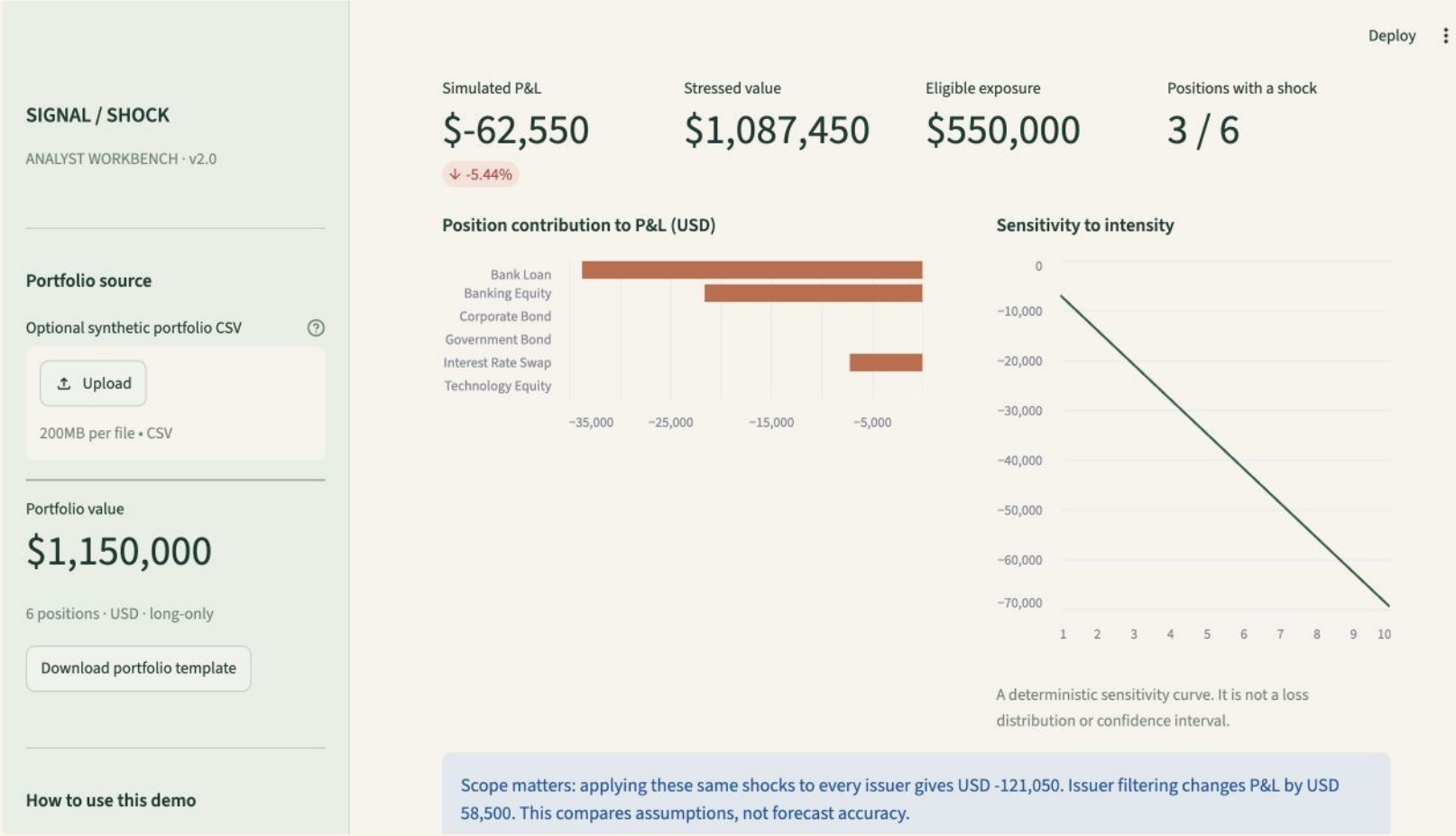
Eligibility mask
Position P&L

AUDIT OUTPUT

Input snapshot + scenario assumptions + position ledger + SHA-256 fingerprint

Python / pandas calculations · Streamlit interface · JSON and CSV exports

A working, auditable prototype



Analyst controls

Upload a synthetic CSV

Inspect matched words

Review ambiguous events

Compare scenarios

Export a complete audit

Issuer scope changes the result

Bank default scenario / intensity 9 / USD 1.15m synthetic portfolio

Bank position	Market value	Applied shock	Simulated P&L
Banking equity	$200,000	−10.8%	−$21,600
Bank loan	$250,000	−13.5%	−$33,750
Interest rate swap	$100,000	−7.2%	−$7,200

−$62,550

3 of 6 positions shocked
Portfolio change: −5.44%

Same shocks applied to every issuer: −$121,050. Scope difference: $58,500.
This compares scenario assumptions, not forecast accuracy.

Evidence and domain relevance

Sentiment on 3,453 sentences	Accuracy	Macro F1
Always-neutral baseline	62.15%	0.2555
Current keyword rules	67.33%	0.4319

Negative recall: 9.52%. Analyst review remains essential.

Historical context

KRE: −19.91% XLF: −2.77% EWU: −5.52%

Selected T−1 to T+5 returns illustrate why event triage matters.
They neither establish causation nor calibrate our shocks.

Limits and the next validation steps

What this prototype establishes

A reproducible headline-to-scenario workflow with issuer filtering, position attribution and explicit assumptions.

What remains unvalidated

Contextual event understanding
Shock calibration and contagion
Bond and derivative repricing

Next experiments

Measure a contextual NLP baseline
Use held-out events for calibration
Add duration and DV01 sensitivities

No trained model or live feed. Synthetic positions. Illustrative scenarios.
The immediate use case is analyst triage and transparent what-if discussion.